

Fiber Propagation Modes

- Main Equations

Considering Cylindrical Coordinates
(Keiser's Book)

Fiber Optic Communications

Optical Fibers

Main Approaches:

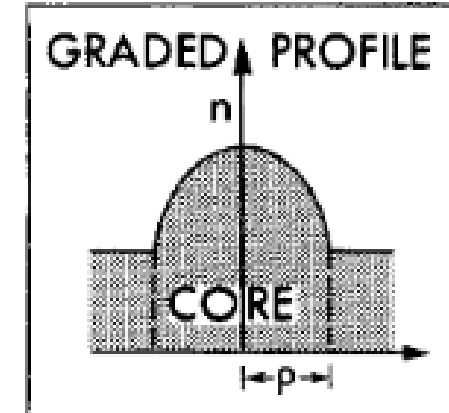
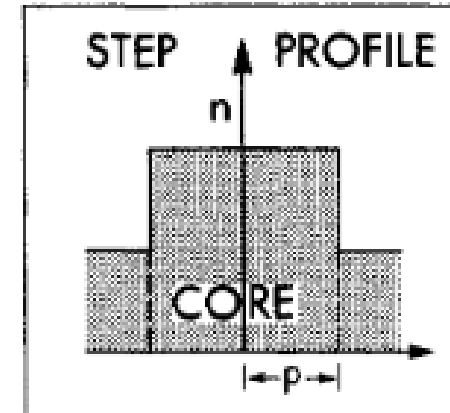
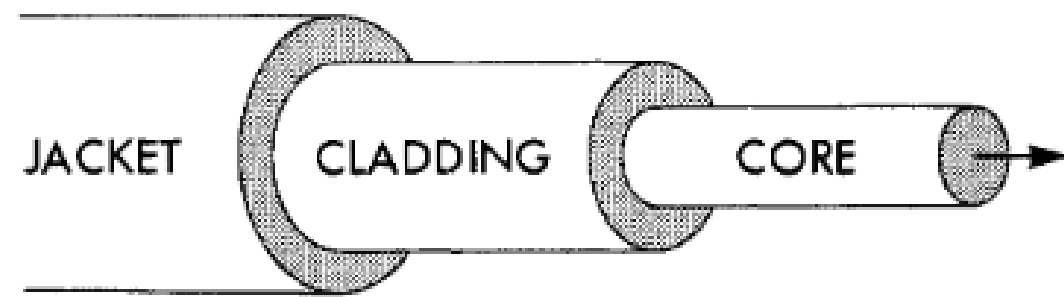
1. Geometric Optics
2. Electromagnetism: Wave Propagation

The Concept of Mode Propagation and formal description

Main Equations:

- Cross-section = modes of irradiance
- **Longitudinal propagation modes**

Silica Fiber



MULTIMODE FIBER

$$12.5 \mu\text{m} < \rho < 100 \mu\text{m}$$

$$0.8 \mu\text{m} < \lambda < 1.6 \mu\text{m}$$

$$0.01 < \Delta < 0.03$$

SINGLE-MODE FIBER

$$2 \mu\text{m} < \rho < 5 \mu\text{m}$$

$$0.8 \mu\text{m} < \lambda < 1.6 \mu\text{m}$$

$$0.003 < \Delta < 0.01$$

Fig. I-1 Nomenclature, profiles and ranges of dimensions for typical optical fibers, where ρ is the core radius, λ is the free-space wavelength of light and $\Delta = (1 - n_d^2/n_{co}^2)/2$ [1].

Wave Equations (obtained from Maxwell's Equations)

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \quad (2-27a)$$

$$\nabla \times \mathbf{H} = \frac{\partial \mathbf{D}}{\partial t} \quad (2-27b)$$

$$\nabla \cdot \mathbf{D} = 0 \quad (2-27c)$$

$$\nabla \cdot \mathbf{B} = 0 \quad (2-27d)$$

Where,

$$\mathbf{D} = \epsilon \mathbf{E} \text{ and } \mathbf{B} = \mu \mathbf{H}.$$

$$\nabla \times (\nabla \times \mathbf{E}) = -\mu \frac{\partial}{\partial t} (\nabla \times \mathbf{H}) = -\epsilon\mu \frac{\partial^2 \mathbf{E}}{\partial t^2} \quad (2-28)$$

Using the vector identity (see App. B)

$$\nabla \times (\nabla \times \mathbf{E}) = \nabla(\nabla \cdot \mathbf{E}) - \nabla^2 \mathbf{E}$$

and using Eq. (2-27c) (that is, $\nabla \cdot \mathbf{E} = 0$), Eq. (2-28) becomes

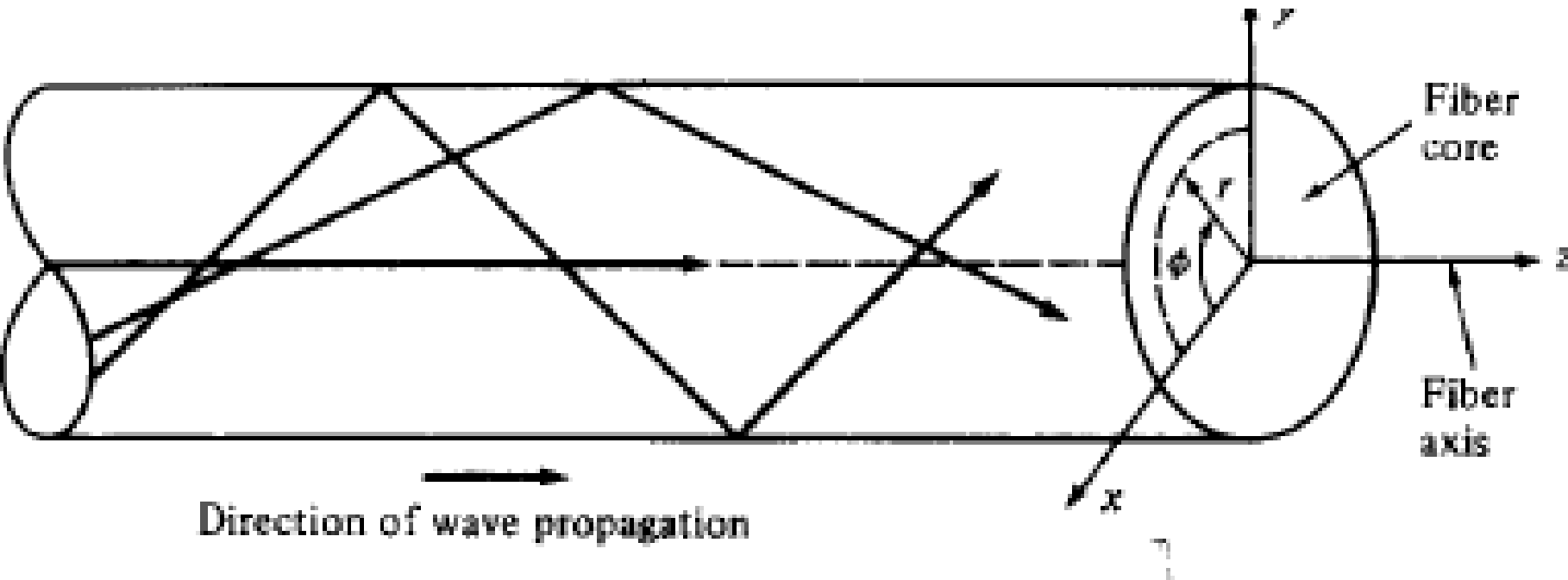
$$\nabla^2 \mathbf{E} = \epsilon\mu \frac{\partial^2 \mathbf{E}}{\partial t^2} \quad (2-29)$$

Similarly, by taking the curl of Eq. (2-27b), it can be shown that

$$\nabla^2 \mathbf{H} = \epsilon\mu \frac{\partial^2 \mathbf{H}}{\partial t^2} \quad (2-30)$$

Equations (2-29) and (2-30) are the standard *wave equations*.

Cylindrical Waveguide



$$\mathbf{E} = \mathbf{E}_0(r, \phi) e^{j(\omega t - \beta z)}$$

$$\mathbf{H} = \mathbf{H}_0(r, \phi) e^{j(\omega t - \beta z)}$$

Substituting
the proposed
solution (E, H)
into the wave
equation

and from Eq. (2-27b)

$$\frac{1}{r} \left(\frac{\partial E_z}{\partial \phi} + jr\beta E_\phi \right) = -j\omega\mu H_r, \quad (2-33a)$$

$$j\beta E_r + \frac{\partial E_z}{\partial r} = j\omega\mu H_\phi \quad (2-33b)$$

$$\frac{1}{r} \left[\frac{\partial}{\partial r} (rE_\phi) - \frac{\partial E_r}{\partial \phi} \right] = -j\mu\omega H_z \quad (2-33c)$$

$$\frac{1}{r} \left(\frac{\partial H_z}{\partial \phi} + jr\beta H_\phi \right) = j\epsilon\omega E_r, \quad (2-34a)$$

$$j\beta H_r + \frac{\partial H_z}{\partial r} = -j\epsilon\omega E_\phi \quad (2-34b)$$

$$\frac{1}{r} \left[\frac{\partial}{\partial r} (rH_\phi) - \frac{\partial H_r}{\partial \phi} \right] = j\epsilon\omega E_z \quad (2-34c)$$

Rewriting eqs as
a function of
 E_z and H_z

$$E_r = -\frac{j}{q^2} \left(\beta \frac{\partial E_z}{\partial r} + \frac{\mu \omega}{r} \frac{\partial H_z}{\partial \phi} \right) \quad (2-35a)$$

$$E_\phi = -\frac{j}{q^2} \left(\frac{\beta}{r} \frac{\partial E_z}{\partial \phi} - \mu \omega \frac{\partial H_z}{\partial r} \right) \quad (2-35b)$$

$$H_r = \frac{-j}{q^2} \left(\beta \frac{\partial H_z}{\partial r} - \frac{\omega \epsilon}{r} \frac{\partial E_z}{\partial \phi} \right) \quad (2-35c)$$

$$H_\phi = \frac{-j}{q^2} \left(\frac{\beta}{r} \frac{\partial H_z}{\partial \phi} + \omega \epsilon \frac{\partial E_z}{\partial r} \right) \quad (2-35d)$$

Where,

$$q^2 = \omega^2 \epsilon \mu - \beta^2 = k^2 - \beta^2$$

Rewriting Equations = Eqs written in Cylindrical Coordinates

Substitution of Eqs. (2-35*c*) and (2-35*d*) into Eq. (2-34*c*) results in the wave equation in cylindrical coordinates

$$\frac{\partial^2 E_z}{\partial r^2} + \frac{1}{r} \frac{\partial E_z}{\partial r} + \frac{1}{r^2} \frac{\partial^2 E_z}{\partial \phi^2} + q^2 E_z = 0 \quad (2-36)$$

and substitution of Eqs. (2-35*a*) and (2-35*b*) into Eq. (2-33*c*) leads to

$$\frac{\partial^2 H_z}{\partial r^2} + \frac{1}{r} \frac{\partial H_z}{\partial r} + \frac{1}{r^2} \frac{\partial^2 H_z}{\partial \phi^2} + q^2 H_z = 0 \quad (2-37)$$

By applying the method of separation of variables

$$E_z = A F_1(r) F_2(\phi) F_3(z) F_4(t) \quad (2-38)$$

Proposing:

$$F_3(z) F_4(t) = e^{j(\omega t - \beta z)} \quad F_2(\phi) = e^{j\nu\phi}$$

We obtain:

$$\frac{\partial^2 F_1}{\partial r^2} + \frac{1}{r} \frac{\partial F_1}{\partial r} + \left(q^2 - \frac{\nu^2}{r^2} \right) F_1 = 0$$

Differential Equations known as Bessel's functions, which solution is given by $J_\nu(ur)$.

Circular
Symmetry
of the
Optical
Fiber

Solutions of Bessel's function

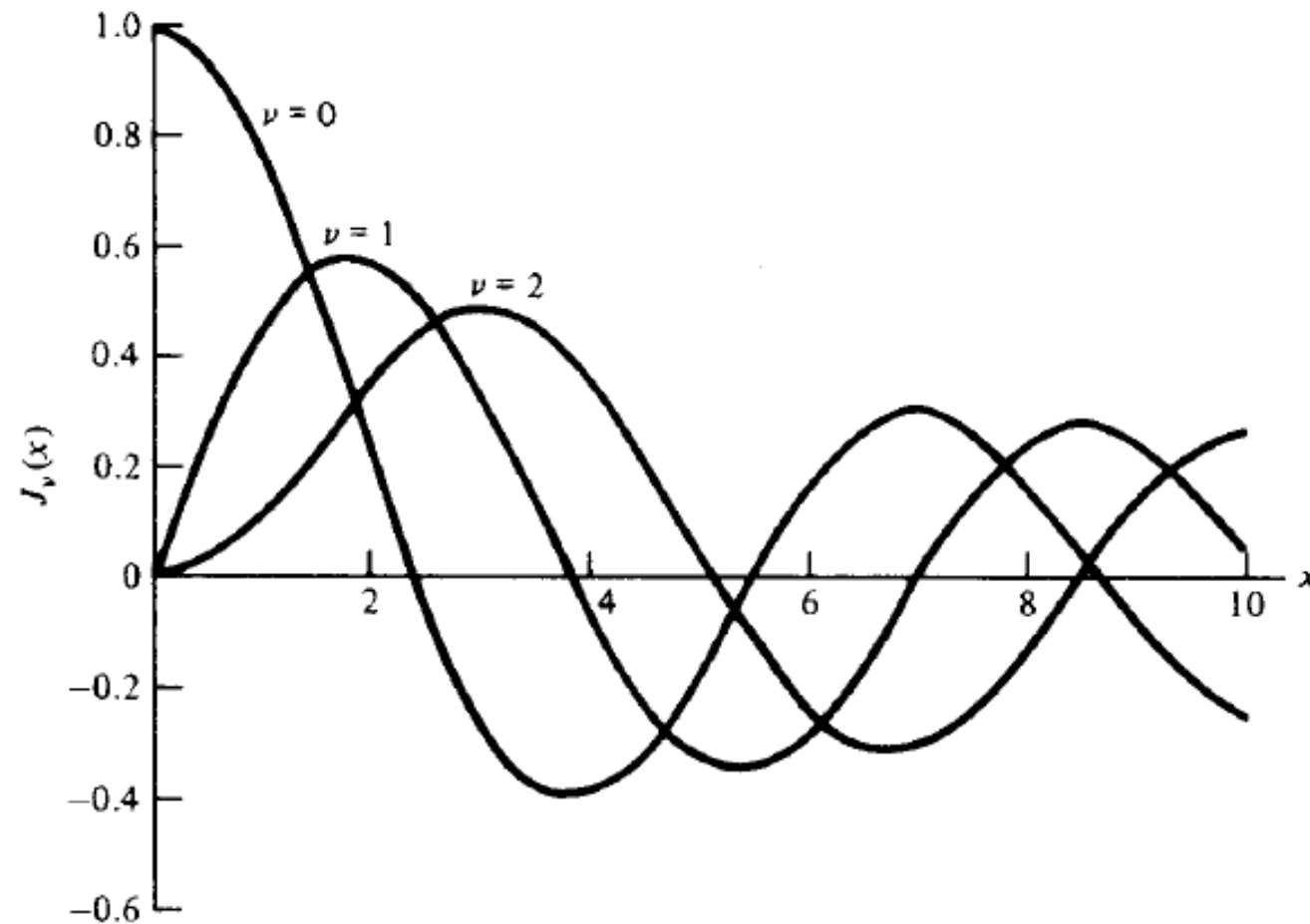


FIGURE 2-16

Variation of the Bessel function $J_\nu(x)$ for the first three orders ($\nu = 0, 1, 2$) plotted as a function of x .

$$E_z(r < a) = AJ_\nu(ur)e^{j\nu\phi}e^{j(\omega t - \beta z)} \quad (2-42)$$

$$H_z(r < a) = BJ_\nu(ur)e^{j\nu\phi}e^{j(\omega t - \beta z)} \quad (2-43)$$

Onde,

$$u^2 = k_1^2 - \beta^2 \quad \text{e} \quad k_1 = 2\pi n_1/\lambda$$

E and H equations

Equations to **E** and **H** outside the core

$$E_z(r > a) = CK_\nu(wr)e^{j\nu\phi}e^{j(\omega t - \beta z)} \quad (2-44)$$

$$H_z(r > a) = DK_\nu(wr)e^{j\nu\phi}e^{j(\omega t - \beta z)} \quad (2-45)$$

Where,

$$k_2 = 2\pi n_2/\lambda$$

After applying the boundary conditions

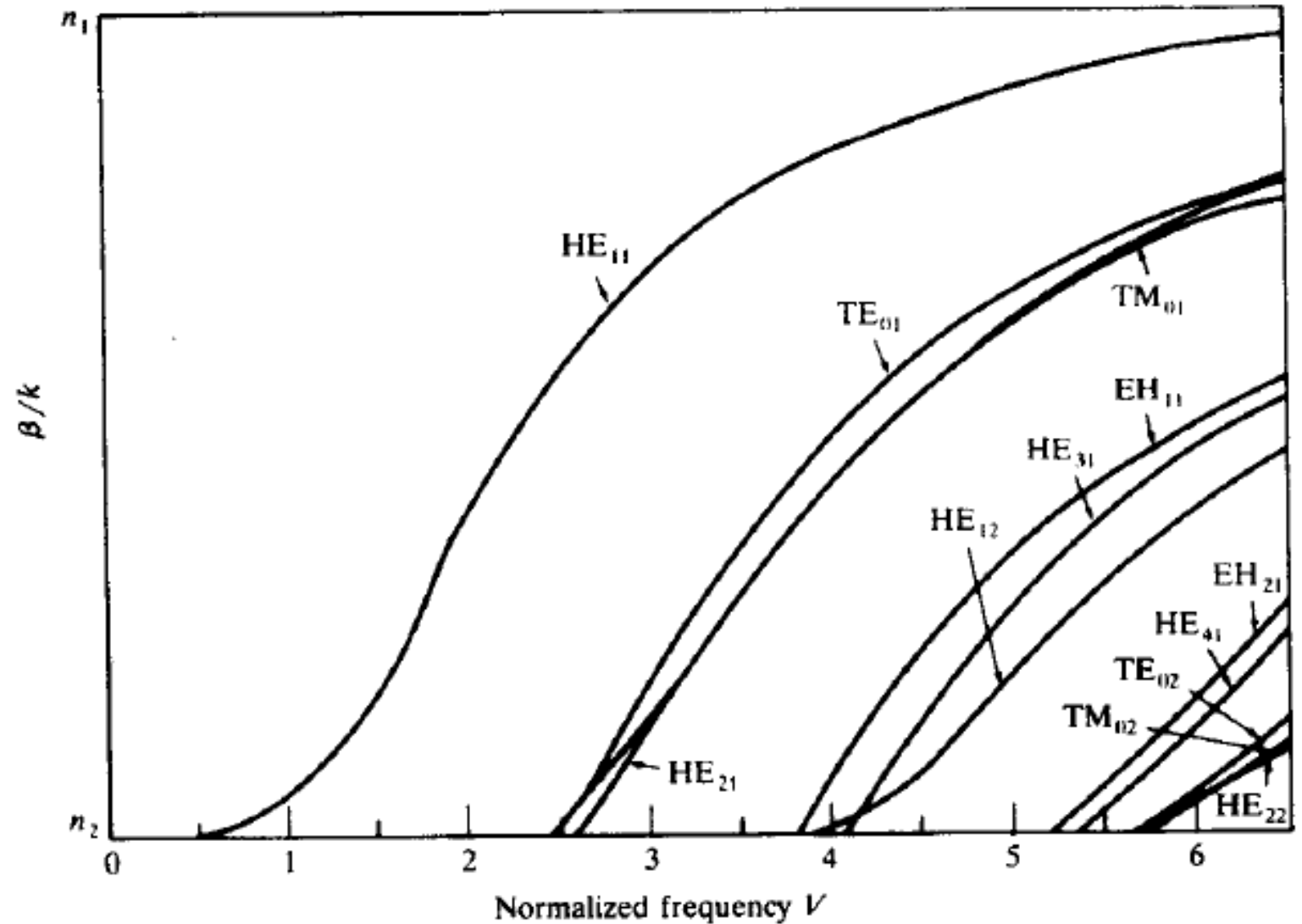
$$n_2k = k_2 \leq \beta \leq k_1 = n_1k$$

Cutoff wavelength λ_c ;

Normalized Frequency V

$$V = \frac{2\pi a}{\lambda} (n_1^2 - n_2^2)^{1/2} \leq 2.405$$

V parameter
and the cutoff
wavelength



2. Optical Properties

- Basic model

Perfect cylindrical symmetry

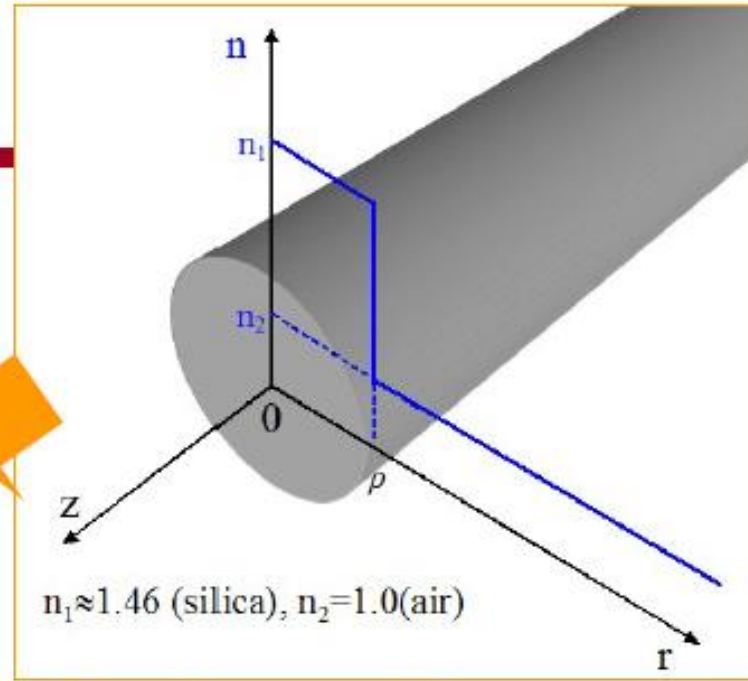


Helmholtz Equations

$$\begin{aligned} (\nabla^2 + n^2 k^2 - \beta^2) \bar{e} &= 0, \\ (\nabla^2 + n^2 k^2 - \beta^2) \bar{h} &= 0. \end{aligned} \quad + \quad n(r) = \begin{cases} n_1, & 0 \leq r < \rho \\ n_2, & \rho \leq r < \infty \end{cases}$$



Analytical solutions of guided modes supported by the fiber [1]



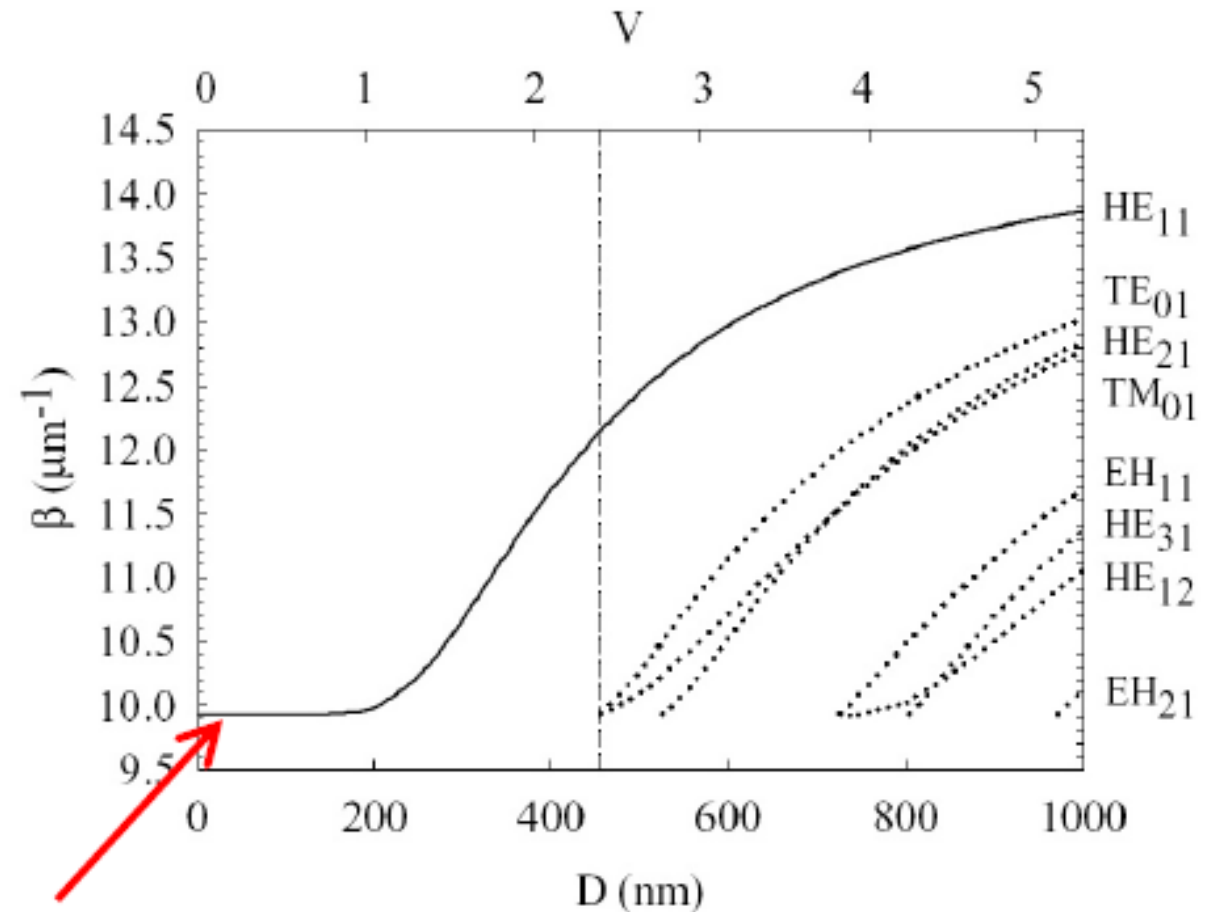
Boundary conditions

Fundamental Equations

Propagation Equation and V Parameter

Propagation constants (β)

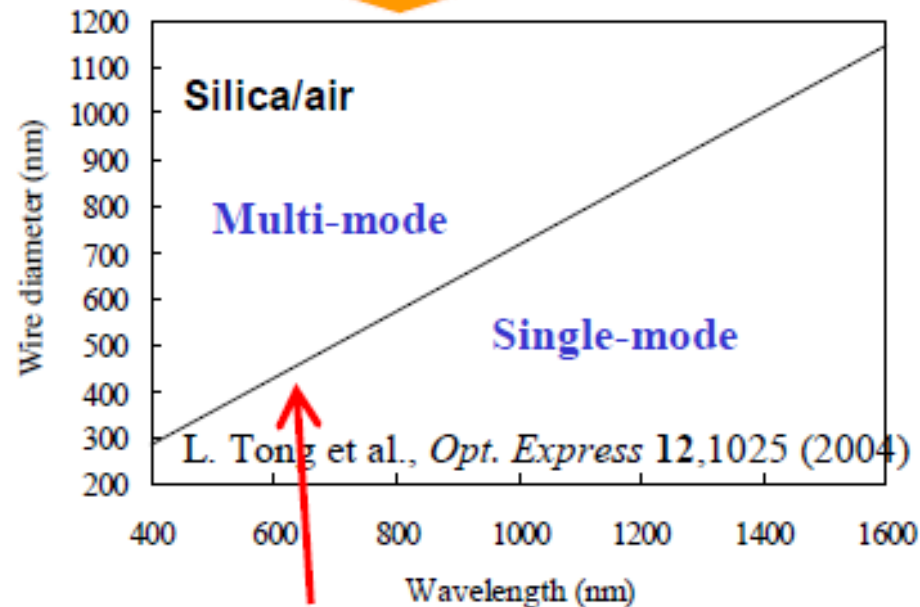
Air-clad silica
microfibers
Wavelength: 633 nm



no cutoff of the fundamental modes

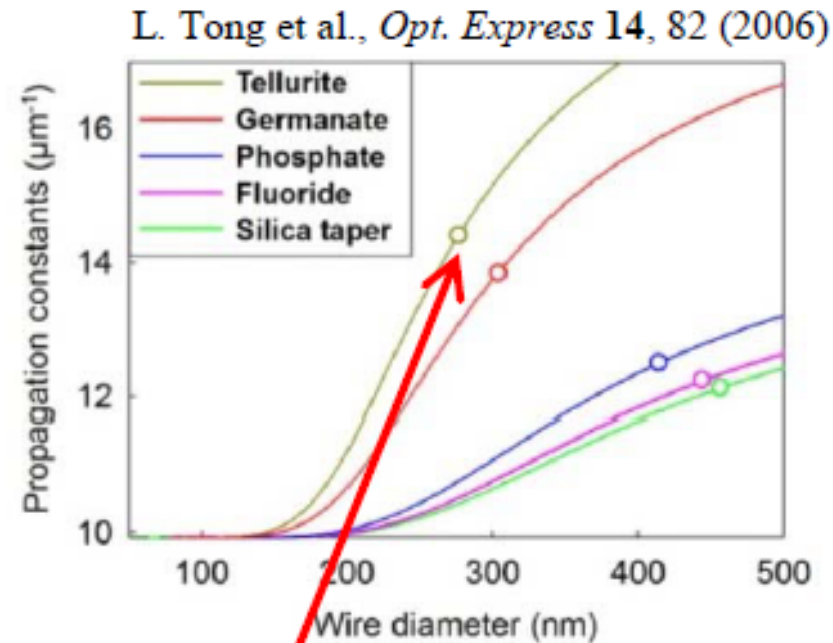
Condition for Singlemode Propagation

$$V = \pi \cdot \frac{D}{\lambda_0} \cdot (n_1^2 - n_2^2)^{1/2} \approx 2.405$$



The shorter the wavelength

β for HE_{11} mode of several glass nanofibers



the higher the refractive index

the smaller the single-mode cutoff diameter

Wave Equation Solution – Main Equations

- Numerically Solving Differential Equations (Eigenvalue equation)



$$\begin{matrix} HE_{vm} \\ EH_{vm} \end{matrix} \left\{ \frac{J'_v(U)}{UJ_v(U)} + \frac{K'_v(W)}{WK_v(W)} \right\} \left\{ \frac{J'_v(U)}{UJ_v(U)} + \frac{n_2^2 K'_v(W)}{n_1^2 WK_v(W)} \right\} = \left(\frac{v\beta}{kn_1} \right)^2 \left(\frac{V}{UW} \right)^4$$

$$TE_{0m} \quad \frac{J_1(U)}{UJ_0(U)} + \frac{K_1(W)}{WK_0(W)} = 0$$

$$TM_{0m} \quad \frac{n_1^2 J_1(U)}{UJ_0(U)} + \frac{n_2^2 K_1(W)}{WK_0(W)} = 0$$

Wave Equation Solution

- **Electric fields of HE_{11} mode**

For the fundamental mode (HE_{11})

Eigenvalue equations

$$\left\{ \frac{J_1'(U)}{UJ_1(U)} + \frac{K_1'(W)}{WK_1(W)} \right\} \left\{ \frac{J_1'(U)}{UJ_1(U)} + \frac{n_2^2 K_1'(W)}{n_1^2 WK_1(W)} \right\} = \left(\frac{\beta}{kn_1} \right)^2 \left(\frac{V}{UW} \right)^4$$



Solve β numerically



Electromagnetic fields $\begin{cases} \vec{E}(r, \phi, z) = (e_r \hat{r} + e_\phi \hat{\phi} + e_z \hat{z}) e^{i\beta z} e^{i\omega t}, \\ \vec{H}(r, \phi, z) = (h_r \hat{r} + h_\phi \hat{\phi} + h_z \hat{z}) e^{i\beta z} e^{i\omega t} \end{cases}$

Proposed Exercise:

Compare the equations implemented on the matlab codes with those discussed on our lecturers (Agrawal e Keiser)

Reference: Keiser's book (Agrawal e Senior)